

Department of CSE

TUTORIAL PLAN

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Branch	Semester	Subject	Code	Session
CSE	3 rd	Data Structures	CCS203	July-Dec.2025

Tutorial Sheet 4

Q1. Convert the following infix expression into postfix using stack.

$(A + B * (C \wedge D - E)) \wedge (F + G * H) - I$

Q2. You are given a sequence of push and pop operations on a stack:

Push(1), Push(2), Push(3), Pop(), Pop(), Push(4), Pop(), Push(5)

- (a) What will be the final stack content?
- (b) Show the state of the stack after every operation.

Q3. The recursive solution of the Towers of Hanoi problem can be simulated using three stacks-one for each rod (A, B, C).

- (a) Explain how each recursive call in the Towers of Hanoi can be replaced by push and pop operations on these stacks.
- (b) Illustrate the process for 3 disks, showing the state of all three stacks after each major move.